

Block 11: complete runtime Solana schema v2 and bounded message signing

1. Identity and frozen scope

Owner: Codex /root, agent-1. Restored worktree: /private/tmp/kk715-stack11-resume.
Branch: audit/715-stack11-agent1-20260927. Target: BitHighlander/keepkey-firmware develop.
Firmware PR: <https://github.com/BitHighlander/keepkey-firmware/pull/877>.
Develop base: fc1e93746132553ad98ed60f4847c8d770732bf9. Report generation predecessor: 3b1d75f0b21623892670be966671be5550669dea.
Reviewed and last hosted-qualified head: f388155ce0b8e8fba1a91e37fe18afaa8f5e2be0.
Earlier local behavior-qualification head: e631449c18f0f26755dc79f59ffea1d6863e8219. Completed block10 predecessor: 476afea2ce9906caaac67bffa900b6a961fa6c4.
Python pin: aae89d378d889b3872696469fe363cf570e5ac4c; companion PR: <https://github.com/BitHighlander/python-keepkey/pull/122>.
Main master-template SHA256 recorded at intake: 4df8723670dbd7266ed8d8756b790725fb4f0f5886d36fe2f9b7c3aece2d64e0.
Main SOP SHA256 recorded at intake: cd2c546d41f0a723f6ce84199a3fb03661f79ab3f49fbafca08de96a7c309e89.

This revision records remediation of review 5333270562 in commit 71e5ea8defa967fc78888c7777749f54906992e4.
Local native and forced-fallback dylib qualification passed after these repairs.
The inventory/PDF are refreshed for this revision; final containing head, final preflight and new hosted results are recorded in the PR receipt after this freeze.

The owner requires the ENTIRE BLOCK PR against fork develop, superseding the older stacked-target instruction for this PR. Develop is an ancestor, so this cumulative integration diff contains accepted predecessor blocks and all block11 source. Their prior audits retain their original scope and limitations. This review also covers the inherited build and scanner defects exposed by that diff.
No merge, assembled-release acceptance or publication is requested.

Block11 includes schema versions 1/2, eight v2 argument slots, token-amount and duration display, same-runtime-signer token metadata, exact instruction coverage, companion restrictions, bounded plain-text raw message signing, AdvancedMode and session trust, and ordinary consent. Certified roots, delegate certificates, LUT proofs and production publication are excluded from runtime 7.15 scope.

Historical PR841 head e97ecd49d03e49536a0fe690bea1319879db022b and base 58526221f47a06d44ef49c9a0642136423f6556f differ in four files despite the retirement comment saying tree-empty. Four core functions are retained from that head in the accepted predecessor; the intake JSON records exact comparisons. The full block's code is visible against develop, including its retained native tests. PR841 is historical; its retirement is not evidence of an empty diff.

2. Git inventory and behavior coverage

Net target inventory command: git diff --numstat fc1e93746132553ad98ed60f4847c8d770732bf9 3b1d75f0b21623892670be966671be5550669dea.
Section 7 lists the complete target inventory from that immutable predecessor.
Counts include code, tests, dependency pointers, workflows, receipts and this report. Historical counts overlap; do not sum them. The remediation removes an unused helper that had been added relative to develop; it is therefore absent from the net target diff. Its test-preservation proof is recorded below.

Behavior	Code-bearing paths in the develop-target PR	Evidence
Schema v1/v2 bounded parser and widths	include/keepkey/firmware/solana.h; lib/firmware/solana.c; unittests/firmware/solana.cpp	Canonical/v2, unsafe-label, trailing-byte and all-prefix truncation assertions; named native cases pass in f388 CI
Exact program/discriminator/data/account binding	lib/firmware/solana.c; unittests/firmware/solana.cpp	Coverage/mismatch/index tests and transfer-companion wire refusal with valid control

Behavior	Code-bearing paths in the develop-target PR	Evidence
Runtime-only trust, token mint and same signer	lib/firmware/fsm_msg_solana.h; lib/firmware/signed_metadata.c; deps/python-keepkey	Runtime token display control and eight invalid-attestation cases; named runtime-attestation test passes in f388 CI
Session reset and policy	lib/firmware/fsm_msg_common.h; lib/firmware/storage.c; deps/python-keepkey	Initialize/ClearSession/policy tests; Solana Initialize refusal
Plain-text predicate and exact message consent	lib/firmware/solana.c; lib/firmware/fsm_msg_solana.h; unittests/firmware/solana.cpp; deps/python-keepkey	Printable/control-byte/key-scan native tests and message-policy wire tests
Malformed input cannot reach schema consent	lib/firmware/fsm_msg_solana.h; deps/python-keepkey	Three malformed forms with/without schema; baseline rejection-order evidence and named runtime-schema regression pass in f388 CI
Transport bounds and host consumer	include/keepkey/transport/messages-solana.options; deps/python-keepkey	raw_tx 2048, schema 256, signature 64, raw message 1024; client/protobuf consumer inspected
Secret detection for tracked dependency paths	.gitleaks.toml	Old configuration misses a tracked-file credential canary; fixed configuration detects it; pinned scanner snapshot/history checks pass
Full/BTC crypto source and test boundaries	deps/crypto/CMakeLists.txt; unittests/crypto/CMakeLists.txt; standalone helper removed from review-head delta	Actual CMake source/command inventories; retained firmware-unit/xunit path; 72 historical passing Zcash cases
Dylib libc fallbacks	lib/emulator/CMakeLists.txt	Forced-missing-libc configure fails before repair and passes after repair; full Linux shared library build/load and string-boundary checks pass
Accepted predecessor stack	Remaining cumulative paths in section 7	Existing accepted reports retained; not attributed to S11-01

Original unit inventories:

Commit a1b63ac5e75c6792f15f8d439e452eca2f2ecb45:

```
13 7 include/keepkey/firmware/solana.h
93 12 lib/firmware/fsm_msg_solana.h
32 7 lib/firmware/solana.c
36 0 unittests/firmware/solana.cpp
```

Commit a7daa79ded5c38afef37c23d02aea6fa671f28a9:

```
6 0 include/keepkey/firmware/solana.h
32 25 lib/firmware/fsm_msg_solana.h
13 0 lib/firmware/solana.c
58 0 unittests/firmware/solana.cpp
```

Dependency pins at qualification:

```
a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec code-signing-keys (heads/master)
8a392f70a5d5575ece3dfb35f115d4a4b27f497c deps/crypto/trezor-firmware (8a392f70a)
5fec9e6906a340be5eb3d795ec746769065b2db8 deps/device-protocol (dice-entropy-v1-83-g5fec9e6)
7888184f28509dba839e3683409443e0b5bb8948 deps/googletest (release-1.8.0-755-g7888184f)
aae89d378d889b3872696469fe363cf570e5ac4c deps/python-keepkey (v6.0.3-958-gaae89d3)
6dfbfdad5d9303ed190d1c3cb7bec34b565b6ce8 deps/qrenc/QR-Code-generator (v1.4.0-11-g6dfbfda)
71d356a1141624994cf613bd2d2583892e8e6d5a deps/sca-hardening/SecAESSTM32 (heads/keepkey)
```

3. Current-source reconciliation and findings

S11-01 FIXED: SolanaSignTx formerly evaluated a valid runtime schema after solana_inspectTx returned MALFORMED with partially populated instructions. That branch preceded the final malformed rejection and skipped signer validation. The early error now zeroes the derived node and returns SyntaxError before schema

evaluation. Baseline emulator tests reached confirmation on all three schema-bearing cases; rejecting the prompt produced ActionCancelled. The fixed handler returns the malformed error without confirmation for all six cases. The baseline test did not approve a signature; signing reachability is source tracing, not a captured baseline signature or a claim of network acceptance.

Security closure: the sensitive operation is Ed25519 signing with the device's derived private key; host input is raw_tx plus a runtime-signed schema. The host observes protocol responses and confirmation requests. Full firmware is affected; BTC does not dispatch Solana. Malformed input must yield no signing consent or signature during parsing/schema selection. Positive controls are valid runtime schema signing and token metadata review. Negative controls are legacy trailing bytes, missing v0 lookup count and a truncated lookup entry, with/without schema. Baseline failures falsify the old guard order; fixed results prove early rejection. The adjacent evidence archive and malformed-schema receipt retain those controls.

Current-source review covers parser exhaustion, safe labels, typed argument bounds and exact byte coverage. Runtime signer ID checks precede narrowing; signatures require exactly 64 bytes and AdvancedMode. Token display requires a matching mint, bounded decimals, valid symbol/signature and the same signer slot as the schema; otherwise it displays base units and mint. Companions cannot silently transfer value; priority fees remain disclosed. Raw printable text rejects the actual signer key at every byte offset and retains message-byte consent. Other message formats retain AdvancedMode.

Review 5333270562, delivered 2026-09-28T02:07:05Z on f388155ce0b8e8fba1a91e37fe18afaa8f5e2be0, reported five findings. Both its overview and all five inline comments were read. The following dispositions describe committed, locally validated changes. Final preflight/hosted CI and published thread dispositions are recorded after freeze.

Finding/comment	Confirmed behavior and local repair	Evidence and qualification
CR11-01 / 4117950392	The inherited ^deps/ scanner exemption hid first-party tracked files as well as dependencies. Removed that exemption and corrected its description; specific public U2F exceptions remain.	Pinned Gitleaks 8.30.1 missed a synthetic credential appended to tracked deps/sca-hardening/aes128_cbc.c with the old configuration and detected it with the fix. A CI-equivalent tracked snapshot and the 219-commit develop..f388 PR range scan clean with the fixed configuration. No new ignore entry was added for this finding.
CR11-02 / 4117950427	Six Pallas/Zcash files were unconditional and five appeared again under KK_ZCASH_PRIVACY. All six, including pallas_ct.c, now belong only to that conditional.	Real CMake evaluation now gives exactly six full-product source entries/compile commands and zero in BTC. The review's duplicate-object consequence was not reproduced: the baseline generator deduplicated 11 entries into six commands. The confirmed defect is the product/source boundary.
CR11-03 / 4117950439	The dylib fallback named strlcpy.c and strlcat.c relative to lib/emulator although they live in lib/board. Both now use explicit board-source paths, with direct CMake boolean checks.	Forcing both KK_HAVE_STRLCPY and KK_HAVE_STRLCAT false made the baseline configure fail with a missing source; repaired host and Linux configurations succeed. Complete GCC Linux dylib build/load and bounded copy/append checks pass with both fallback flags forced false.

Finding/comment	Confirmed behavior and local repair	Evidence and qualification
CR11-04 / 4117950452	The separate zcash-crypto-unit target existed in BTC builds and relied on the unconditional crypto sources. Removed that redundant target and its now-unreferenced deterministic RNG helper.	CMake inventories confirm no standalone target in either product, retained conditional firmware-unit Zcash coverage in full, and no Zcash implementation source in BTC (shared generated protocol metadata remains). No unique test implementation was removed.
CR11-05 / 4117950469	The standalone target was absent from xunit, but the assertion that these tests never ran is refuted. The same firmware/zcash.cpp suite already belongs to firmware-unit and runs through xunit/CTest. Removed the redundant target and documented the existing registration.	Exact reviewed-head CI contains 72 passing Zcash cases in full firmware.xml and zero in BTC. The repaired source registration retains that suite. The repaired full xunit run again passes all 72 Zcash cases; BTC passes 180 native tests with none of these cases.

The underlying classes are broad path-based scanner exemptions, product flags applied inconsistently across library/test targets, and optional host branches whose relative source paths were never exercised. Controls inspect a real tracked path, generated source/compile inventories and forced capability-false configure branches. Test preservation is established from actual JUnit and the existing required execution target, rather than executable names alone.

These controls and the complete review are preserved in the adjacent 715-11-review1-controls-20260927.tgz archive with a verified SHA256 manifest. Local disposition records are /private/tmp/kk715-stack11-review-crypto/disposition.json (with before.json and after.json), /private/tmp/kk715-stack11-review-fixes/secret-scan/result.json, and the baseline-host.log, fixed-host.log and fixed-linux-configure.log files in /private/tmp/kk715-stack11-review-fixes/dylib. Review body/comments are retained in /private/tmp/kk715-stack11-evidence-resume/reviews-final.json and review-inline-final.json. The extra initialized-submodule scanner diagnostic reports 72 fixture/public-material matches across 23 third-party files. They were classified separately; that diagnostic is not a passing qualification scan and no exemption was added. CI's scanner checkout does not initialize those gitlinks.

CI-TARGET-01: the develop-based PR event still fails closed because develop has no KK_RELEASE_MISSING_CAPABILITIES ledger. The owner approved the independent manual authority KK_ACCEPTED_WAIVER_SHA=2483977523d9b6beab52a69429ace9aaef00f76; its seven-capability ledger contains the candidate's five waivers. That repository setting remains active. Manual run 36363905103 qualified f388 with publishing disabled, without changing runtime authority selection or waiving the PR check. A new manual run is required for the current build/scanner remediation.

CI-MANUAL-02 FIXED: run 36363232771 failed because a PR-authority test inherited workflow_dispatch and the approved variable. The fixture explicitly sets pull_request; the old control fails and the fixed report-gate suite passes in that environment. CI-MANUAL-03 FIXED: two public dependency SHA lines in a block09 receipt needed exact no-git path/rule/line fingerprints in addition to existing commit-scoped fingerprints. git ls-tree proves their gitlink identities; a same-file credential canary still triggers the pinned scanner. Both repairs were included in successful f388 CI. They are distinct from CR11-01's inherited broad exemption, which is now removed locally.

4. Verification and artifact identity

The following hosted results are exact for the reviewed head f388155ce0b8e8fba1a91e37fe18afaa8f5e2be0, not the current post-review remediation. Run <https://github.com/BitHighlander/keepkey-firmware/actions/runs/36363905103> completed successfully, including the release-evidence gate; publishing was skipped.

Evidence	Verified result on f388
Combined authoritative report	1,521 pass, 84 skip, zero fail/error; 1,605 total
Full native	firmware 669, board 19, crypto 18 pass; includes 72 Zcash cases

Evidence	Verified result on f388
Full Python integration	811 pass, 84 skip; both new runtime-schema regression methods pass
Dylib integration	4 pass
BTC native	firmware 143, board 19, crypto 18 pass; Solana/Zcash suites excluded
BTC Python integration	347 pass, 548 skip; new Solana methods explicitly skip because full firmware is required
Dedicated contracts, full	Stack06 13 pass/3 skip; Stack07 21 pass; Stack09 1 pass; Stack10 7 pass
OLED evidence	1,347 frame hashes and 194 sequence manifests verified; screenshot selection 194 pass/24 skip
ARM variants	Full and BTC manifests, firmware/Python identities and all 23 listed binary/ELF hashes per variant verified
SRAM gates	Full reserve 17,800 B; BTC 39,136 B; largest frame 7,664 B in each; both pass

The combined count excludes BTC and repeated contract/capture runs. Named passing cases are Solana.SchemaParsesCanonicalPayload, Solana.SchemaV2ParsesTokenDurationAndEightArgs, tests.test_msg_solana_schema_v2.TestSolanaSchemaRuntime.test_runtime_schema_cannot_rescue_malformed_transaction, and tests.test_msg_solana_schema_v2.TestSolanaSchemaRuntime.test_runtime_schema_requires_complete_valid_runtime_attestation. Their JUnit identities and statuses were checked directly.

Report artifact 10946638964; full ARM artifact 10946334201; BTC ARM artifact 10947200608.
 Report PDF SHA256: 3166384b3568ef0b3716c825a531a35b853c3708c0f23d2110b1c16effdd2165.
 Report manifest SHA256: 5bbdb05076248c6586d5a008d56a1696209167b886c3a8f5080616c89fd56827.
 Merged JUnit SHA256: 7cd8e675ba9777d6042ab68f0f738e9b2e13fc029e148b84869e31d0ec8d85c7.
 All nine JUnit input hashes, generator hashes, ARM hashes and OLED hashes agree.
 The manual event intentionally has an empty firmware_pr metadata field; the receipt binds PR877 to the exact head/run. Python metadata names PR122 and pin aae89d378d889b3872696469fe363cf570e5ac4c. Independent verification files are retained under /private/tmp/kk715-stack11-ci-verification, including final-verified.json, run-final.json, jobs-final.json and test-report/test-report-manifest.json.

Earlier local evidence on e631449c remains historical: 669 firmware-unit passes; 39 Solana/runtime/session wire passes plus 14 passing subtests and three skips. The S11-01 baseline and fixed controls remain in the adjacent evidence archive. Restoration of the unavailable original temporary worktree verified all 11 archived artifact hashes before publishing the reviewed candidate.

Material skips remain explicit: certified/attestor tests require firmware 7.16; LUT proofs are outside runtime 7.15 scope. The two session power-cycle tests skip because the harness does not own the UDP emulator process (hosted run port 11044; earlier local run port 12144). Actual power cycles, physical-device/OLED behavior and release promotion remain separate gates. The complete 84-skip ledger is in the verified report manifest; none is represented as a passing test.

Current remediation qualification on 71e5ea8defa967fc78888c7777749f54906992e4: full native xunit passes 706 tests (669 firmware, 19 board, 18 crypto), including all 72 Zcash cases; BTC xunit passes 180 tests with no Zcash implementation units or symbols in the tested archives/binaries. Both have zero failures/skips. Forced-libc Linux dylib configure/build/load and truncation/zero-capacity copy and append checks pass using pinned CMake 3.31.6 and GCC 6.4.0 in image7438e53933d4. The old bundled Clang lacks stdatomic.h; that unsuccessful optional-dylib attempt is preserved alongside the successful GCC build. Missing nested token data was initialized at its pinned commit before both successful native builds. Final preflight, new exact-head hosted CI and artifact verification are recorded in the PR receipt after this report freeze; f388 results remain historical.

5. Report render and local preflight

This source and its PDF reconcile the delivered review and repair validation. Section 7 uses the explicit immutable base-to-predecessor inventory. Final Git path/count equality, PDF legibility and post-edit preflight are verified before push; their receipt identifies the resulting containing head. The earlier report/PDF and preflight remain evidence of the reviewed checkpoint only.

The entire develop-target diff is code-bearing, with block11 behavior mapped

above and a pinned Python companion. The pre-push gate must account for removal of the unused standalone RNG helper while proving retained Zcash registration. Record the final containing head, complete validation results and all published finding dispositions in the PR receipt without rewriting historical evidence.

6. Copilot checkpoint and completion

Owner authorization: exactly one request. Requests used: 1/1. Timeline event 31957245566 registered that request at 2026-09-28T01:58:42Z. Review 5333270562 was delivered at 2026-09-28T02:07:05Z against f388155ce0b8e8fba1a91e37fe18afaa8f5e2be0, with Lite effort and five findings (overview: four high, one medium). Its verdict is Changes recommended, not approval. It has been fully inspected; the five local dispositions are recorded in section 3.

After the remediation commit, that review remains historical and does not cover the new head. Local fixes, verification, publication and addressed-thread closure can proceed within the authorized work; no second Copilot request is authorized. Published replies and thread resolution are pending this final validation/push. Do not report Copilot-clean status or block11 completion from a stale review or from test totals. Predecessor decisions, outstanding release gates and the separate develop-event authority incompatibility retain their stated status.

7. Complete target diff inventory

Added	Deleted	Path
1	2	.gitattributes
755	587	.github/workflows/ci.yml
8	8	.github/workflows/mirror-base-image.yml
275	69	.github/workflows/release.yml
6	0	.gitignore
20	0	.gitleaks.toml
9	0	.gitleaksignore
3	3	.gitmodules
58	4	CMakeLists.txt
1	0	cmake/caches/device.cmake
40	0	cmake/toolchains/mingw-w64-x86_64.cmake
13	0	deps/crypto/CMakeLists.txt
1	1	deps/crypto/trezor-firmware
1	1	deps/device-protocol
1	1	deps/python-keepkey
192	0	docs/DiceEntropy.md
154	0	docs/dice-vs-coldcard.md
94	0	docs/firmware/reviews/7.15.0-review-round2-remediation.md
124	0	docs/release/7.14.3-COMBINED-CANDIDATE.md
230	0	docs/release/REHEARSAL-SOP.md
148	0	docs/release/RELEASE-PROGRAM.md
63	0	docs/release/audit-units/715-04-review-20260923.md
-	-	docs/release/audit-units/715-04-review-20260923.pdf
61	0	docs/release/audit-units/715-05-review-20260924.md

Added	Deleted	Path
-	-	docs/release/audit-units/715-05-review-20260924.pdf
678	0	docs/release/audit-units/715-05-skips-20260924.tsv
20	0	docs/release/audit-units/715-06-evidence-sha256-20260924.txt
82	0	docs/release/audit-units/715-06-history-20260924.tsv
-	-	docs/release/audit-units/715-06-intake-evidence-20260924.tgz
140	0	docs/release/audit-units/715-06-reconciliation-20260924.md
185	0	docs/release/audit-units/715-06-review-20260924.md
-	-	docs/release/audit-units/715-06-review-20260924.pdf
663	0	docs/release/audit-units/715-06-skips-20260924.tsv
-	-	docs/release/audit-units/715-06-validation-20260924.tgz
1086	0	docs/release/audit-units/715-06-validation-sha256-20260924.txt
144	0	docs/release/audit-units/715-07-first-copilot-retrospective-20260924.md
-	-	docs/release/audit-units/715-07-first-copilot-retrospective-20260924.pdf
81	0	docs/release/audit-units/715-07-full-audit-remediation-20260924.md
-	-	docs/release/audit-units/715-07-full-audit-remediation-20260924.pdf
112	0	docs/release/audit-units/715-07-history-20260924.tsv
62	0	docs/release/audit-units/715-07-intake-20260924.md
141	0	docs/release/audit-units/715-07-review-20260924.md
-	-	docs/release/audit-units/715-07-review-20260924.pdf
577	0	docs/release/audit-units/715-07-skips-20260924.tsv
-	-	docs/release/audit-units/715-07-validation-20260924.tgz
112	0	docs/release/audit-units/715-07-validation-sha256-20260924.txt
42	0	docs/release/audit-units/715-07b-d-resplit-mapping-20260927.md
35	0	docs/release/audit-units/715-07b-resplit-checks-20260927.json
74	0	docs/release/audit-units/715-07b-resplit-receipt-20260927.md
128	0	docs/release/audit-units/715-07b1-formatters-audit-20260926.md
-	-	docs/release/audit-units/715-07b1-formatters-audit-20260926.pdf
119	0	docs/release/audit-units/715-07b2a-groups-audit-20260927.md
-	-	docs/release/audit-units/715-07b2a-groups-audit-20260927.pdf
121	0	docs/release/audit-units/715-07b2b-iteration-audit-20260927.md
-	-	docs/release/audit-units/715-07b2b-iteration-audit-20260927.pdf
136	0	docs/release/audit-units/715-07b3r-embedded-audit-20260927.md
-	-	docs/release/audit-units/715-07b3r-embedded-audit-20260927.pdf
203	0	docs/release/audit-units/715-08-10-integration-20260927.md
-	-	docs/release/audit-units/715-08-10-integration-20260927.pdf
-	-	docs/release/audit-units/715-08-10-local-evidence-20260927.tgz
110	0	docs/release/audit-units/715-08-audit-20260927.md
-	-	docs/release/audit-units/715-08-audit-20260927.pdf
86	0	docs/release/audit-units/715-08-consolidated-audit-20260927.md
-	-	docs/release/audit-units/715-08-consolidated-audit-20260927.pdf
-	-	docs/release/audit-units/715-08-local-evidence-20260927.tgz

Added	Deleted	Path
175	0	docs/release/audit-units/715-08-reconciliation-20260927.json
80	0	docs/release/audit-units/715-08B-audit-20260927.md
-	-	docs/release/audit-units/715-08B-audit-20260927.pdf
114	0	docs/release/audit-units/715-09-audit-20260927.md
-	-	docs/release/audit-units/715-09-audit-20260927.pdf
-	-	docs/release/audit-units/715-09-local-evidence-20260927.tgz
145	0	docs/release/audit-units/715-09-reconciliation-20260927.json
163	0	docs/release/audit-units/715-10-audit-20260927.md
-	-	docs/release/audit-units/715-10-audit-20260927.pdf
31	0	docs/release/audit-units/715-10-completion-plan-20260927.md
-	-	docs/release/audit-units/715-10-final-readiness-evidence-20260927.tgz
73	0	docs/release/audit-units/715-10-intake-20260927.md
48	0	docs/release/audit-units/715-10-reconciliation-20260927.json
639	0	docs/release/audit-units/715-11-complete-block-audit-20260927.md
-	-	docs/release/audit-units/715-11-complete-block-audit-20260927.pdf
200	0	docs/release/audit-units/715-11-intake-20260927.json
36	0	docs/release/audit-units/715-11-intake-20260927.md
-	-	docs/release/audit-units/715-11-local-evidence-20260927.tgz
82	0	docs/release/audit-units/715-11-malformed-schema.md
-	-	docs/release/audit-units/715-11-manual-ci-controls-20260927.tgz
-	-	docs/release/audit-units/715-11-review1-controls-20260927.tgz
26	0	docs/release/audit-units/P01-002-artifact-routing.md
27	0	docs/release/audit-units/P01-003-firmware-packaging.md
12	0	docs/release/audit-units/P01-004-report-inputs.md
16	0	docs/release/audit-units/P01-008-compose-variant.md
27	0	docs/release/audit-units/P02-001-tiny-lifetime.md
14	0	docs/release/audit-units/P02-003-packet-lifetime.md
18	0	docs/release/audit-units/P06-002-ripple-display-response.md
15	0	docs/release/audit-units/P06-003-ripple-display-test.md
21	0	docs/release/audit-units/P06-005-ripple-unsupported-memo.md
12	0	docs/release/audit-units/P06-006-ripple-memo-rejection-test.md
23	0	docs/release/audit-units/decode-boundaries.md
15	0	docs/release/audit-units/eos-authorization.md
87	0	docs/release/audit-units/full-diff-audit-20260912.md
14	0	docs/release/audit-units/passphrase-transition.md
32	0	docs/release/audit-units/scope-repair.md
14	0	docs/release/audit-units/source-storage-policy.md
21	0	docs/release/audit-units/storage-capacities.md
11	0	docs/release/audit-units/storage-cipher-cleanup.md
11	0	docs/release/audit-units/storage-revocation.md
6	0	docs/release/audit-units/token-build-dependencies.md

Added	Deleted	Path
170	0	docs/security/7.14.3-bitcoin-only-dice-audit-sop.md
235	0	docs/security/HANDOFF-ERC7730-715-FORMATTERS.md
295	0	docs/security/HANDOFF-ERC7730-ALPHA.md
209	0	docs/security/HANDOFF-ERC7730-PHASE-E.md
-	-	docs/security/evidence/rom-printf-integer-percent/01-thorchain-withdraw-25.05pct.png
-	-	docs/security/evidence/rom-printf-integer-percent/02-thorchain-sending-eth.png
17	0	docs/security/evidence/rom-printf-integer-percent/README.md
28	0	include/keepkey/board/bsd_compat.h
67	1	include/keepkey/board/confirm_sm.h
35	0	include/keepkey/board/draw.h
16	0	include/keepkey/board/keepkey_display.h
21	0	include/keepkey/board/layout.h
57	24	include/keepkey/board/messages.h
5	0	include/keepkey/board/usb.h
61	17	include/keepkey/emulator/libkemu.h
43	1	include/keepkey/firmware/app_confirm.h
9	1	include/keepkey/firmware/app_layout.h
11	1	include/keepkey/firmware/authenticator.h
12	0	include/keepkey/firmware/binance.h
22	0	include/keepkey/firmware/bip85.h
2	0	include/keepkey/firmware/coins.def
3	1	include/keepkey/firmware/coins.h
72	0	include/keepkey/firmware/dice_input.h
11	4	include/keepkey/firmware/eip712.h
227	0	include/keepkey/firmware/eip712_stream.h
4	0	include/keepkey/firmware/eos.h
80	0	include/keepkey/firmware/erc7730_abi.h
90	0	include/keepkey/firmware/erc7730_abi_stream.h
111	0	include/keepkey/firmware/erc7730_capabilities.h
241	0	include/keepkey/firmware/erc7730_catalog.h
20	0	include/keepkey/firmware/erc7730_condition.h
69	0	include/keepkey/firmware/erc7730_field.h
38	0	include/keepkey/firmware/erc7730_format.h
311	0	include/keepkey/firmware/erc7730_program.h
26	0	include/keepkey/firmware/erc7730_tx.h
277	0	include/keepkey/firmware/erc7730_workflow.h
20	2	include/keepkey/firmware/ethereum.h
34	0	include/keepkey/firmware/ethereum_contracts.h
3	0	include/keepkey/firmware/ethereum_contracts/saproxy.h
27	1	include/keepkey/firmware/ethereum_contracts/thortx.h
3	0	include/keepkey/firmware/ethereum_contracts/zxliquidtx.h

Added	Deleted	Path
14	3	include/keepkey/firmware/ethereum_tokens.h
38	0	include/keepkey/firmware/fsm.h
98	0	include/keepkey/firmware/hive.h
2	0	include/keepkey/firmware/home_sm.h
54	4	include/keepkey/firmware/mayachain.h
36	1	include/keepkey/firmware/osmosis.h
7	0	include/keepkey/firmware/pin_sm.h
9	0	include/keepkey/firmware/recovery_cipher.h
65	4	include/keepkey/firmware/reset.h
28	2	include/keepkey/firmware/ripple.h
242	0	include/keepkey/firmware/signed_metadata.h
21	1	include/keepkey/firmware/signing.h
16	3	include/keepkey/firmware/signtx_tendermint.h
148	0	include/keepkey/firmware/solana.h
45	1	include/keepkey/firmware/storage.h
30	0	include/keepkey/firmware/tendermint.h
50	5	include/keepkey/firmware/thorchain.h
4	1	include/keepkey/firmware/tiny-json.h
0	7	include/keepkey/firmware/tokens.def
10	1	include/keepkey/firmware/transaction.h
56	0	include/keepkey/firmware/tron.h
3	1	include/keepkey/firmware/txin_check.h
431	0	include/keepkey/firmware/zcash.h
19	0	include/keepkey/rand/rng.h
143	0	include/keepkey/rand/rng_health.h
2	0	include/keepkey/transport/interface.h
27	0	include/keepkey/transport/messages-ethereum.options
56	0	include/keepkey/transport/messages-hive.options
0	1	include/keepkey/transport/messages-osmosis.options
2	0	include/keepkey/transport/messages-ripple.options
12	0	include/keepkey/transport/messages-solana.options
1	0	include/keepkey/transport/messages-thorchain.options
54	0	include/keepkey/transport/messages-zcash.options
16	0	include/keepkey/transport/messages.options
222	113	include/pb.h
512	30	lib/board/confirm_sm.c
163	16	lib/board/draw.c
4	1	lib/board/font.c
13	10	lib/board/keepkey_board.c
16	6	lib/board/keepkey_flash.c
152	9	lib/board/layout.c

Added	Deleted	Path
162	68	lib/board/messages.c
39	9	lib/board/signatures.c
28	2	lib/board/timer.c
3	0	lib/board/udp.c
50	23	lib/board/usb.c
48	16	lib/board/util.c
11	4	lib/emulator/CMakeLists.txt
302	35	lib/emulator/libkemu.c
45	5	lib/emulator/setup.c
8	2	lib/emulator/udp.c
56	27	lib/firmware/CMakeLists.txt
281	44	lib/firmware/app_confirm.c
117	4	lib/firmware/app_layout.c
240	131	lib/firmware/authenticator.c
141	14	lib/firmware/binance.c
105	0	lib/firmware/bip85.c
19	1	lib/firmware/coins.c
448	0	lib/firmware/dice_input.c
460	255	lib/firmware/eip712.c
1456	0	lib/firmware/eip712_stream.c
21	8	lib/firmware/eos-contracts/eosio.system.c
117	13	lib/firmware/eos.c
472	0	lib/firmware/erc7730_abi.c
463	0	lib/firmware/erc7730_abi_stream.c
243	0	lib/firmware/erc7730_capabilities.c
1652	0	lib/firmware/erc7730_catalog.c
109	0	lib/firmware/erc7730_condition.c
297	0	lib/firmware/erc7730_field.c
233	0	lib/firmware/erc7730_format.c
877	0	lib/firmware/erc7730_program.c
46	0	lib/firmware/erc7730_tx.c
1097	0	lib/firmware/erc7730_workflow.c
659	119	lib/firmware/ethereum.c
51	11	lib/firmware/ethereum_contracts.c
36	24	lib/firmware/ethereum_contracts/makerdao.c
52	13	lib/firmware/ethereum_contracts/saproxy.c
189	47	lib/firmware/ethereum_contracts/thortx.c
113	83	lib/firmware/ethereum_contracts/zxappliquid.c
154	111	lib/firmware/ethereum_contracts/zxliquidtx.c
201	36	lib/firmware/ethereum_contracts/zxswap.c
80	16	lib/firmware/ethereum_contracts/zxtransERC20.c

Added	Deleted	Path
21	13	lib/firmware/ethereum_tokens.c
416	19	lib/firmware/fsm.c
58	17	lib/firmware/fsm_msg_binance.h
143	0	lib/firmware/fsm_msg_bip85.h
52	9	lib/firmware/fsm_msg_coin.h
143	35	lib/firmware/fsm_msg_common.h
118	23	lib/firmware/fsm_msg_cosmos.h
27	14	lib/firmware/fsm_msg_crypto.h
24	7	lib/firmware/fsm_msg_debug.h
22	3	lib/firmware/fsm_msg_eos.h
2239	141	lib/firmware/fsm_msg_ethereum.h
491	0	lib/firmware/fsm_msg_hive.h
162	30	lib/firmware/fsm_msg_mayachain.h
210	113	lib/firmware/fsm_msg_osmosis.h
81	9	lib/firmware/fsm_msg_ripple.h
814	153	lib/firmware/fsm_msg_solana.h
125	23	lib/firmware/fsm_msg_tendermint.h
136	29	lib/firmware/fsm_msg_thorchain.h
37	55	lib/firmware/fsm_msg_ton.h
111	15	lib/firmware/fsm_msg_tron.h
1699	0	lib/firmware/fsm_msg_zcash.h
439	0	lib/firmware/hive.c
46	8	lib/firmware/home_sm.c
389	97	lib/firmware/mayachain.c
54	0	lib/firmware/messagemap.def
5	2	lib/firmware/nano.c
270	37	lib/firmware/osmosis.c
77	6	lib/firmware/passphrase_sm.c
47	21	lib/firmware/pin_sm.c
213	90	lib/firmware/recovery_cipher.c
482	96	lib/firmware/reset.c
65	10	lib/firmware/ripple.c
1046	0	lib/firmware/signed_metadata.c
549	59	lib/firmware/signing.c
193	65	lib/firmware/signtx_tendermint.c
538	41	lib/firmware/solana.c
306	42	lib/firmware/storage.c
8	0	lib/firmware/storage.h
180	4	lib/firmware/tendermint.c
318	46	lib/firmware/thorchain.c
5	3	lib/firmware/tiny-json.c

Added	Deleted	Path
5	3	lib/firmware/ton.c
180	18	lib/firmware/transaction.c
364	1	lib/firmware/tron.c
14	7	lib/firmware/txin_check.c
11	2	lib/firmware/u2f.c
1237	0	lib/firmware/zcash.c
2	1	lib/rand/CMakeLists.txt
97	13	lib/rand/rng.c
348	0	lib/rand/rng_health.c
18	0	lib/transport/CMakeLists.txt
2	1	lib/transport/pb_decode.c
1	1	scripts/build/docker/device/debug.sh
7	2	scripts/build/docker/device/release.sh
1	1	scripts/build/docker/emulator/debug.sh
1	0	scripts/cppcheck-version
32	0	scripts/cppcheck.sh
10	2	scripts/emulator/Dockerfile
111	0	scripts/emulator/capture-dice-flow.py
99	0	scripts/emulator/capture-thor-percent.py
12	0	scripts/emulator/docker-compose.bitcoin-only.yml
14	0	scripts/emulator/docker-compose.yml
19	1	scripts/emulator/firmware-unit.sh
87	65	scripts/emulator/python-keepkey-tests.sh
1	1	scripts/emulator/python-keepkey.Dockerfile
443	0	scripts/emulator/test_stack07_regressions.py
36	0	scripts/emulator/test_stack09_integration.py
141	0	scripts/emulator/test_stack10_regressions.py
623	68	scripts/generate-test-report.py
103	0	scripts/preflight.sh
92	0	scripts/preflight_checks.py
249	0	scripts/test_generate_test_report.py
124	0	scripts/test_preflight_checks.py
91	0	scripts/test_preflight_cppcheck.py
85	0	scripts/tests/test_usb_receive_callbacks.py
36	0	scripts/verify-token-def.py
1	0	tools/CMakeLists.txt
124	0	tools/check_sram_budget.py
33	11	tools/emulator/CMakeLists.txt
20	0	tools/erc7730-validate/CMakeLists.txt
76	0	tools/erc7730-validate/main.c
3	1	tools/firmware/CMakeLists.txt

Added	Deleted	Path
10	0	tools/firmware/keepkey.ld
14	9	tools/firmware/keepkey_main.c
10	0	tools/sram-budgets.json
251	0	tools/verify_dice_seed.py
1	1	unittests/board/CMakeLists.txt
490	2	unittests/board/board.cpp
6	2	unittests/crypto/CMakeLists.txt
500	0	unittests/crypto/bip340.cpp
55	7	unittests/firmware/CMakeLists.txt
48	0	unittests/firmware/app_confirm.cpp
302	0	unittests/firmware/authenticator.cpp
151	0	unittests/firmware/binance.cpp
28	2	unittests/firmware/coins.cpp
141	0	unittests/firmware/confirm_test_utils.cpp
166	8	unittests/firmware/cosmos.cpp
116	0	unittests/firmware/dice.cpp
194	0	unittests/firmware/eip712.cpp
1076	0	unittests/firmware/eip712_stream.cpp
90	3	unittests/firmware/eos.cpp
50	0	unittests/firmware/eos_authorization.cpp
16	0	unittests/firmware/eos_authorization_probe.c
174	0	unittests/firmware/erc7730_abi.cpp
371	0	unittests/firmware/erc7730_abi_stream.cpp
1901	0	unittests/firmware/erc7730_catalog.cpp
105	0	unittests/firmware/erc7730_condition.cpp
304	0	unittests/firmware/erc7730_field.cpp
224	0	unittests/firmware/erc7730_format.cpp
411	0	unittests/firmware/erc7730_program.cpp
74	0	unittests/firmware/erc7730_tx.cpp
207	0	unittests/firmware/erc7730_workflow.cpp
709	1	unittests/firmware/ethereum.cpp
1204	0	unittests/firmware/fsm.cpp
783	0	unittests/firmware/hive.cpp
20	0	unittests/firmware/kkconfirm_driver.h
54	0	unittests/firmware/liquidity_key_probe.c
390	7	unittests/firmware/mayachain.cpp
11	0	unittests/firmware/messages_probe.c
35	0	unittests/firmware/nanopb_bounds.cpp
153	0	unittests/firmware/osmosis.cpp
46	0	unittests/firmware/recovery.cpp
37	0	unittests/firmware/ripple.cpp

Added	Deleted	Path
297	0	unittests/firmware/rng_health.cpp
193	0	unittests/firmware/setup_ceremony.cpp
1804	0	unittests/firmware/signed_metadata.cpp
143	0	unittests/firmware/signing.cpp
969	15	unittests/firmware/solana.cpp
316	29	unittests/firmware/storage.cpp
129	0	unittests/firmware/storage_cipher_probe.c
24	0	unittests/firmware/storage_cipher_probe.h
140	0	unittests/firmware/storage_passphrase.cpp
28	0	unittests/firmware/test_board.cpp
350	21	unittests/firmware/thorchain.cpp
173	0	unittests/firmware/transaction.cpp
604	0	unittests/firmware/tron.cpp
169	0	unittests/firmware/usb_rx.cpp
2204	0	unittests/firmware/zcash.cpp